ICS 253 – Discrete Structures
Chapter 5: Number Theory

Question bank: 90 multiple choice + 10 written problems

Questions from ICS 253 old quizzes and homework (AI_Work). **[new]** marks new questions written in the same style.

Part A: Multiple choice

Q1. [new]

Which of the following statements is true?

(A) $3 \mid 12$

(B) $12 \mid 3$

(C) $0 \mid 5$

(D) $4 \mid 10$

Answer: (A). $12 = 4 \cdot 3$. $3/12$ is not an integer, 0 never divides anything, and $10 = 2.5 \cdot 4$.

My answer

Q2. [new]

Using the definition "$x \mid y$ if and only if $x \neq 0$ and $y = kx$ for some integer k", which statement is true?

(A) $0 \mid 0$

(B) $7 \mid 0$

(C) $0 \mid 7$

(D) $7 \mid 1$

Answer: (B). $0 = 0 \cdot 7$ and $7 \neq 0$. Any statement with 0 on the left fails because the divisor must be nonzero.

My answer

Q3. [new]

Which of the following statements is *false*?

(A) $-3 \mid 12$

(B) $3 \mid -12$

(C) $6 \mid -18$

(D) $-4 \mid 18$

Answer: (D). $18/(-4) = -4.5$ is not an integer. The others hold: $12 = (-4)(-3)$, $-12 = (-4)(3)$, $-18 = (-3)(6)$.

My answer

Q4. [new]

How many positive divisors does 36 have?

(A) 6

(B) 8

(C) 9

(D) 12

Answer: (C). $1, 2, 3, 4, 6, 9, 12, 18, 36$.

My answer

Q5. [new]

The set of all positive divisors of 28 is

(A) $\{2, 4, 7, 14\}$

(B) $\{1, 2, 4, 7, 14\}$

(C) $\{1, 2, 3, 4, 7, 14, 28\}$

(D) $\{1, 2, 4, 7, 14, 28\}$

Answer: (D). A number is always a divisor of itself, and 1 divides every integer. $3 \nmid 28$.

My answer

Q6. [new]

Let a and b be integers with $7 \mid a$ and $7 \mid b$. Which of the following must be divisible by 7?

(A) $a + 1$

(B) $3a - 5b$

(C) $2a + 5$

(D) $ab + 1$

Answer: (B). By the theorem on linear combinations, if $7 \mid a$ and $7 \mid b$ then $7 \mid (sa + tb)$ for all integers s, t; take $s = 3$, $t = -5$.

My answer

Q7. [new]

Since $6 \mid 30$ and $6 \mid 42$, the number 6 divides every linear combination of 30 and 42. Which of the following numbers is such a linear combination and is therefore divisible by 6?

(A) $30 + 42 + 1$

(B) $30 \cdot 2 + 5$

(C) $30 \cdot 4 - 42 \cdot 2$

(D) $42 - 30 + 4$

Answer: (C). $30 \cdot 4 - 42 \cdot 2 = 36 = 6 \cdot 6$. The others are 73, 65, 16, none of which is a linear combination of 30 and 42.

My answer

Q8. [new]

What are $45 \operatorname{div} 7$ and $45 \bmod 7$?

(A) 6 and 3

(B) 7 and -4

(C) 6 and 4

(D) 5 and 10

Answer: (A). $45 = 6 \cdot 7 + 3$ with $0 \leq 3 \leq 6$.

My answer

Q9. [new]

According to the Division Algorithm, what are $-29 \operatorname{div} 7$ and $-29 \bmod 7$?

(A) -4 and -1

(B) -4 and 1

(C) -5 and -6

(D) -5 and 6

Answer: (D). $-29 = (-5) \cdot 7 + 6$ and the remainder must lie in $\{0, \ldots, 6\}$.

My answer

Q10. [new]

A programming language returns -2 when computing $-17 \bmod 5$. What is the value of $-17 \bmod 5$ according to the Division Algorithm?

(A) -2

(B) 2

(C) 3

(D) -3

Answer: (C). The remainder must be in $\{0, 1, 2, 3, 4\}$: $-17 = (-4) \cdot 5 + 3$.

My answer

Q11. [new]

What is $-1 \bmod 8$?

(A) -1

(B) 7

(C) 1

(D) 0

Answer: (B). $-1 = (-1) \cdot 8 + 7$.

My answer

Q12. [new]

What is $-40 \operatorname{div} 6$?

(A) -6

(B) 6

(C) -7

(D) -8

Answer: (C). $-40 = (-7) \cdot 6 + 2$ with $0 \leq 2 \leq 5$. (-6 would give remainder -4, which is not allowed.)

My answer

Q13. [new]

Which of the following integer division statements is true?

(A) $-23 \bmod 6 = -5$

(B) $-23 \operatorname{div} 6 = -3$

(C) $-23 \bmod 6 = 1$

(D) $23 \operatorname{div} 6 = 4$

Answer: (C). $-23 = (-4) \cdot 6 + 1$, so $-23 \bmod 6 = 1$ and $-23 \operatorname{div} 6 = -4$; also $23 \operatorname{div} 6 = 3$.

My answer

Q14. [new]

Find n if $n \operatorname{div} 7 = 9$ and $n \bmod 7 = 4$.

(A) 67

(B) 63

(C) 36

(D) 71

Answer: (A). $n = 9 \cdot 7 + 4 = 67$.

My answer

Q15. [new]

Find n if $n \operatorname{div} 5 = -6$ and $n \bmod 5 = 2$.

(A) -32

(B) 28

(C) -28

(D) -13

Answer: (C). $n = (-6) \cdot 5 + 2 = -28$.

My answer

Q16. [new]

In which of the following sets does *every* element x satisfy $x \bmod 4 = 3$?

(A) $\{3, -3, 11\}$

(B) $\{15, -1, 7\}$

(C) $\{5, 9, -5\}$

(D) $\{-3, 1, 13\}$

Answer: (B). $15 = 3 \cdot 4 + 3$, $-1 = (-1) \cdot 4 + 3$, $7 = 4 + 3$. Note $-3 \bmod 4 = 1$.

My answer

Q17. [new]

Let d be a positive integer and $0 \le n < d$. Then $(n \operatorname{div} d,\ n \bmod d)$ equals

(A) $(1, 0)$

(B) $(n, 0)$

(C) $(0, n)$

(D) $(0, d - n)$

Answer: (C). $n = 0 \cdot d + n$ and $0 \le n \le d - 1$, so by uniqueness $q = 0$, $r = n$.

My answer

Q18. [new]

Which statement is true for all integers a, b, c (with the divisors nonzero)?

(A) If $a \mid b$ and $a \mid c$, then $b \mid c$.

(B) If $a \mid bc$, then $a \mid b$.

(C) If $a \mid (b + c)$, then $a \mid b$.

(D) If $a \mid b$ and $b \mid c$, then $a \mid c$.

Answer: (D). $b = ka$, $c = jb = (jk)a$. Counterexamples for the others: $2 \mid 4$, $2 \mid 6$, $4 \nmid 6$; $6 \mid 2 \cdot 3$, $6 \nmid 2$; $2 \mid 1 + 1$, $2 \nmid 1$.

My answer

Q19. [new]

Which choice of (a, b, c) is a counterexample to the statement "for all integers a, b, c, if $a \mid bc$ then $a \mid b$ or $a \mid c$"?

(A) $a = 3,\ b = 6,\ c = 1$

(B) $a = 6,\ b = 2,\ c = 3$

(C) $a = 5,\ b = 10,\ c = 2$

(D) $a = 2,\ b = 3,\ c = 5$

Answer: (B). $6 \mid 6 = 2 \cdot 3$ but $6 \nmid 2$ and $6 \nmid 3$. In (A) and (C) the conclusion holds; in (D) the hypothesis $2 \mid 15$ fails.

My answer

Q20. [new]

The Division Algorithm states: if n is an integer and d is a positive integer, there are unique integers q and r with _____ such that $n = qd + r$. Fill in the blank.

(A) $0 \le r \le d - 1$

(B) $0 < r \le d$

(C) $-d < r < d$

(D) $0 \le r \le d$

Answer: (A). The remainder $n \bmod d$ always lies in $\{0, 1, \ldots, d-1\}$.

My answer

Q21. [new]

The procedural Division Algorithm for $n < 0$ is: $q := 0$; $r := n$; While $(r < 0)$: $r := r + d$; $q := q - 1$; End-While. On input $n = -25$, $d = 7$, how many times does the loop execute, and what is the output?

(A) 3 times; $q = -3$, $r = -4$

(B) 4 times; $q = -4$, $r = 3$

(C) 4 times; $q = 4$, $r = 3$

(D) 3 times; $q = -3$, $r = 4$

Answer: (B). $-25 \to -18 \to -11 \to -4 \to 3$: four additions of 7, so $q = -4$, $r = 3$; check $(-4) \cdot 7 + 3 = -25$.

My answer

Q22. [new]

Which number can *not* be written as $4s + 6t$ for integers s and t?

(A) 2

(B) 10

(C) -8

(D) 7

Answer: (D). $2 \mid 4$ and $2 \mid 6$, so 2 divides every linear combination $4s + 6t$; 7 is odd. ($2 = 4 \cdot 2 - 6$, $10 = 4 + 6$, $-8 = -2 \cdot 4$.)

My answer

Q23. [new]

How many integers n with $1 \le n \le 100$ satisfy $n \bmod 7 = 3$?

(A) 13

(B) 14

(C) 15

(D) 7

Answer: (B). They are $3, 10, 17, \ldots, 94$, i.e. $3 + 7k$ for $k = 0, \ldots, 13$.

My answer

Q24. [new]

The statement $5 \mid 35$ means that

(A) 5 is a multiple of 35

(B) 35 is a multiple of 5

(C) $35/5$ is not an integer

(D) $5 = 35k$ for some integer k

Answer: (B). $35 = 7 \cdot 5$; so 35 is a multiple of 5 and 5 is a factor (divisor) of 35.

My answer

Q25. [new]

Which statement is true for all integers a, b, c with $a \neq 0$?

(A) If $a \mid b$, then $a \mid bc$.

(B) If $a \mid (b + c)$, then $a \mid b$ or $a \mid c$.

(C) If $a \mid b$, then $b \mid a$.

(D) If $a \mid bc$, then $a \mid b$.

Answer: (A). If $b = ka$ then $bc = (kc)a$. Counterexamples: $2 \mid 1 + 1$; $2 \mid 4$ but $4 \nmid 2$; $6 \mid 2 \cdot 3$.

My answer

Q26. [new]

Which of the following integer division statements is true?

(A) $17 \bmod 3 = 2$ and $17 \operatorname{div} 3 = 5$

(B) $-17 \bmod 3 = -2$

(C) $-17 \operatorname{div} 3 = -5$

(D) $-17 \bmod 3 = 2$

Answer: (A). $17 = 5 \cdot 3 + 2$. For -17: $-17 = (-6) \cdot 3 + 1$, so $-17 \bmod 3 = 1$ and $-17 \operatorname{div} 3 = -6$.

My answer

Q27. Quiz 7 Alenzi

Find the multiplicative inverse of $719 \pmod{311}$. (Your answer must be a positive integer.)

(A) 109

(B) 202

(C) 252

(D) 97

Answer: (B). The extended Euclidean algorithm gives $1 = (-109) \cdot 719 + 252 \cdot 311$, so $719 \cdot (-109) \equiv 1 \pmod{311}$ and $-109 \equiv 311 - 109 = 202$.

My answer

Q28.

Quiz 7 Alenzi

Find the multiplicative inverse of $311 \pmod{719}$. (Your answer must be a positive integer.)

(A) 467

(B) 109

(C) 202

(D) 252

Answer: (D). From $1 = (-109) \cdot 719 + 252 \cdot 311$, reducing mod 719 gives $252 \cdot 311 \equiv 1 \pmod{719}$.

My answer

Q29.

[new]

The GCD Theorem, on which Euclid's algorithm is based, states that for positive integers x and y,

(A) $\gcd(x, y) = \gcd(y,\ x \bmod y)$

(B) $\gcd(x, y) = \gcd(y,\ x \operatorname{div} y)$

(C) $\gcd(x, y) = x \bmod y$

(D) $\gcd(x, y) = \gcd(x \bmod y,\ x \operatorname{div} y)$

Answer: (A). Any common divisor of x and y divides $x \bmod y = x - qy$, and conversely any common divisor of y and $x \bmod y$ divides $x = qy + (x \bmod y)$.

My answer

Q30.

[new]

What is $\gcd(84, 36)$?

(A) 6

(B) 4

(C) 12

(D) 18

Answer: (C). $84 \bmod 36 = 12$, $36 \bmod 12 = 0$, so the gcd is 12.

My answer

Q31. [new]

What is $\gcd(252, 198)$?

(A) 9

(B) 18

(C) 36

(D) 6

Answer: (B). $252 \bmod 198 = 54$, $198 \bmod 54 = 36$, $54 \bmod 36 = 18$, $36 \bmod 18 = 0$.

My answer

Q32. [new]

Which is the sequence of numbers generated by Euclid's algorithm on inputs 330 and 156 (each new number is the previous-but-one mod the previous one)?

(A) $330, 156, 18, 6, 0$

(B) $330, 156, 24, 12, 0$

(C) $330, 156, 18, 12, 6, 0$

(D) $330, 156, 18, 12, 0$

Answer: (C). $330 \bmod 156 = 18$, $156 \bmod 18 = 12$, $18 \bmod 12 = 6$, $12 \bmod 6 = 0$; so $\gcd(330, 156) = 6$.

My answer

Q33. [new]

Euclid's algorithm (zyBook form): If $(y > x)$ swap x and y; $r := x \bmod y$; While $(r \neq 0)$: $x := y$; $y := r$; $r := x \bmod y$; End-while; Return(y). How many times does the loop execute on inputs $x = 330$, $y = 156$?

(A) 2

(B) 3

(C) 4

(D) 5

Answer: (B). Before the loop $r = 18$. Iterations: $(x, y, r) = (156, 18, 12)$, $(18, 12, 6)$, $(12, 6, 0)$; then the loop stops and 6 is returned.

My answer

Q34. [new]

Running the Euclid's algorithm of the previous question on $x = 330$, $y = 156$, what is the value of r after the second iteration of the loop?

(A) 12

(B) 18

(C) 6

(D) 0

Answer: (C). After iteration 1: $x = 156$, $y = 18$, $r = 12$. After iteration 2: $x = 18$, $y = 12$, $r = 6$.

My answer

Q35. [new]

Euclid's algorithm (as above) is run on inputs 1001 and 91. What does it return, and how many times does the loop execute?

(A) 91; the loop executes 0 times

(B) 7; the loop executes 1 time

(C) 13; the loop executes 2 times

(D) 91; the loop executes 1 time

Answer: (A). $1001 = 11 \cdot 91$, so $r = 1001 \bmod 91 = 0$ before the loop; the loop is skipped and $y = 91$ is returned.

My answer

Q36. [new]

Using the extended Euclidean algorithm, find integers s and t with $6 = 330s + 156t$.

(A) $s = -19,\ t = 9$

(B) $s = 9,\ t = -19$

(C) $s = 8,\ t = -17$

(D) $s = 1,\ t = -2$

Answer: (B). $18 = 330 - 2 \cdot 156$, $12 = 156 - 8 \cdot 18$, $6 = 18 - 12$. So $6 = 18 - (156 - 8 \cdot 18) = 9 \cdot 18 - 156 = 9 \cdot 330 - 19 \cdot 156$.

My answer

Q37. [new]

Given $6 = 9 \cdot 330 - 19 \cdot 156$, which of the following is another pair (s, t) with $6 = 330s + 156t$?

(A) $(35, -45)$

(B) $(22, -47)$

(C) $(35, -74)$

(D) $(9, 19)$

Answer: (C). Adding $156/6 = 26$ to s and subtracting $330/6 = 55$ from t keeps the sum: $35 \cdot 330 - 74 \cdot 156 = 11550 - 11544 = 6$.

My answer

Q38. [new]

Euclid's algorithm on 161 and 28 gives $161 = 5 \cdot 28 + 21$, $28 = 1 \cdot 21 + 7$, $21 = 3 \cdot 7$. Which expresses $\gcd(161, 28)$ as $161s + 28t$?

(A) $7 = (-1) \cdot 161 + 6 \cdot 28$

(B) $7 = 1 \cdot 161 - 6 \cdot 28$

(C) $7 = 6 \cdot 161 - 1 \cdot 28$

(D) $7 = (-1) \cdot 161 + 5 \cdot 28$

Answer: (A). $7 = 28 - 21 = 28 - (161 - 5 \cdot 28) = 6 \cdot 28 - 161$.

My answer

Q39. [new]

Euclid's algorithm on 360 and 84 generates the sequence $360, 84, 24, 12$, where $24 = 360 - B \cdot 84$ and $12 = 84 - A \cdot 24$. What are A and B?

(A) $A = 4,\ B = 3$

(B) $A = 3,\ B = 3$

(C) $A = 4,\ B = 4$

(D) $A = 3,\ B = 4$

Answer: (D). $360 \operatorname{div} 84 = 4$ ($360 - 336 = 24$) and $84 \operatorname{div} 24 = 3$ ($84 - 72 = 12$).

My answer

Q40. [new]

Continuing the previous question, $\gcd(360, 84) = 12 = 360s + 84t$ where

(A) $s = 3,\ t = -13$

(B) $s = -3,\ t = 13$

(C) $s = 13,\ t = -3$

(D) $s = -4,\ t = 17$

Answer: (B). $12 = 84 - 3(360 - 4 \cdot 84) = 13 \cdot 84 - 3 \cdot 360$.

My answer

Q41. [new]

By inspection, the multiplicative inverse of $3 \bmod 7$ is

(A) 2

(B) 4

(C) 3

(D) 5

Answer: (D). $3 \cdot 5 = 15 = 2 \cdot 7 + 1$.

My answer

Q42. [new]

What is the multiplicative inverse of $5 \bmod 12$?

(A) 5

(B) 7

(C) 11

(D) 2

Answer: (A). $5 \cdot 5 = 25 = 2 \cdot 12 + 1$, so 5 is its own inverse mod 12.

My answer

Q43. [new]

Which of the following integers does *not* have a multiplicative inverse mod 10?

(A) 3

(B) 7

(C) 4

(D) 9

Answer: (C). An inverse mod n exists iff $\gcd(x, n) = 1$; $\gcd(4, 10) = 2$. (Inverses: $3 \cdot 7 \equiv 1$, $9 \cdot 9 \equiv 1$.)

My answer

Q44. [new]

What is the multiplicative inverse of $7 \bmod 26$?

(A) 11

(B) 15

(C) 19

(D) 3

Answer: (B). $7 \cdot 15 = 105 = 4 \cdot 26 + 1$.

My answer

Q45. [new]

Euclid's extended algorithm returns $\gcd(40, 17) = 1 = 3 \cdot 40 - 7 \cdot 17$. What is the multiplicative inverse of $17 \bmod 40$ (an integer from 1 through 39)?

(A) 7

(B) 3

(C) 37

(D) 33

Answer: (D). The coefficient of 17 is -7, and $-7 \bmod 40 = 33$. Check: $17 \cdot 33 = 561 = 14 \cdot 40 + 1$.

My answer

Q46. [new]

Euclid's extended algorithm returns $\gcd(57, 23) = 1 = 5 \cdot 23 - 2 \cdot 57$. What is the multiplicative inverse of $57 \bmod 23$ (an integer from 1 through 22)?

(A) 2

(B) 5

(C) 21

(D) 18

Answer: (C). The coefficient of 57 is $-2 \equiv 21 \pmod{23}$. Check: $57 \cdot 21 = 1197 = 52 \cdot 23 + 1$.

My answer

Q47. [new]

For integers x and $n > 1$, x has a multiplicative inverse mod n if and only if

(A) $x < n$

(B) n is prime

(C) $\gcd(x, n) = 1$

(D) $x \mid n$

Answer: (C). If $\gcd(x, n) = 1 = sx + tn$, then $xs \equiv 1 \pmod{n}$; if $\gcd(x, n) = d > 1$ then every $xy \bmod n$ is a multiple of d, never 1.

My answer

Q48. [new]

How many elements of $\{1, 2, \ldots, 11\}$ have a multiplicative inverse mod 12?

(A) 4

(B) 5

(C) 6

(D) 11

Answer: (A). Exactly those relatively prime to 12: $1, 5, 7, 11$.

My answer

Q49. [new]

Suppose $\gcd(x, y) = d$ for positive integers x and y. What is $\gcd(x/d,\ y/d)$?

(A) d

(B) 1

(C) d^2

(D) xy/d

Answer: (B). If $c > 1$ divided both x/d and y/d, then cd would be a common divisor of x and y larger than d.

My answer

Q50. [new]

Since $12 \cdot 3 + 35 \cdot (-1) = 1$, what is $\gcd(12, 35)$?

(A) 12

(B) 35

(C) 1

(D) It cannot be determined from this equation.

Answer: (C). $\gcd(12, 35)$ divides 12 and 35, hence divides $12 \cdot 3 + 35 \cdot (-1) = 1$; so it equals 1.

My answer

Q51. [new]

Why is Euclid's algorithm, rather than prime factorization, used to compute gcds in cryptographic protocols?

(A) Prime factorization gives the wrong gcd for large numbers.

(B) No efficient method is known to factor numbers with hundreds of digits, but Euclid's algorithm is efficient.

(C) Euclid's algorithm only works for prime numbers.

(D) Euclid's algorithm avoids division.

Answer: (B). Euclid's algorithm needs only repeated mod operations, while factoring huge numbers is believed to be hard.

My answer

Q52. [new]

Using the extended Euclidean algorithm, the multiplicative inverse of $17 \bmod 43$ is

(A) 5

(B) 38

(C) 23

(D) 12

Answer: (B). $43 = 2 \cdot 17 + 9$, $17 = 9 + 8$, $9 = 8 + 1$; so $1 = 9 - 8 = 2 \cdot 9 - 17 = 2 \cdot 43 - 5 \cdot 17$, and $-5 \equiv 38$. Check: $17 \cdot 38 = 646 = 15 \cdot 43 + 1$.

My answer

Q53. Quiz 7 Alenzi

In $\mathbb{Z}_4$, what is $3 + 7$?

(A) 10

(B) 3

(C) 2

(D) 0

Answer: (C). $3 + 7 = 10 = 4 \cdot 2 + 2$, so $10 \bmod 4 = 2$.

My answer

Q54.

Quiz 7 Alenzi

In $\mathbb{Z}_5$, what is $12 - 3$?

(A) 9

(B) 4

(C) 1

(D) 3

Answer: (B). $12 - 3 = 9 = 5 \cdot 1 + 4$, so the result is 4.

My answer

Q55.

Quiz 7 Alenzi

In $\mathbb{Z}_6$, what is 5×11?

(A) 55

(B) 5

(C) 4

(D) 1

Answer: (D). $5 \cdot 11 = 55 = 6 \cdot 9 + 1$.

My answer

Q56.

[new]

What is the value of 5 times 6 in $\mathbb{Z}_7$?

(A) 30

(B) 2

(C) 4

(D) 1

Answer: (B). $30 = 4 \cdot 7 + 2$.

My answer

Q57. [new]

What is the value of 7 plus 8 in $\mathbb{Z}_9$?

(A) 15

(B) 7

(C) 5

(D) 6

Answer: (D). $15 = 9 + 6$.

My answer

Q58. [new]

Which pair of integers is congruent mod 5?

(A) 17 and -3

(B) 17 and 3

(C) -3 and 3

(D) 12 and -12

Answer: (A). $17 \bmod 5 = 2$ and $-3 \bmod 5 = 2$ (equivalently, $5 \mid 17 - (-3) = 20$).

My answer

Q59. [new]

Let $m > 1$. By the alternate characterization of congruence, $x \equiv y \pmod{m}$ if and only if

(A) $m \mid (x + y)$

(B) $m \mid (x - y)$

(C) $x \mid (y - m)$

(D) $(x - y) \mid m$

Answer: (B). $x \bmod m = y \bmod m$ exactly when $x - y$ is a multiple of m.

My answer

Q60. [new]

Which of the following is congruent to $23 \pmod{6}$?

(A) 3

(B) -5

(C) -1

(D) 14

Answer: (C). $23 \bmod 6 = 5$ and $-1 \bmod 6 = 5$. ($-5 \bmod 6 = 1$, $14 \bmod 6 = 2$.)

My answer

Q61. [new]

Using the convention that mod has the same precedence as multiplication, evaluate $17 + 12 \bmod 5$.

(A) 4

(B) 19

(C) 2

(D) 29

Answer: (B). mod is applied before $+$: $17 + (12 \bmod 5) = 17 + 2 = 19$.

My answer

Q62. [new]

Evaluate $4 \cdot 9 \bmod 7$ (operations of equal precedence are performed left to right).

(A) 8

(B) 36

(C) 1

(D) 2

Answer: (C). $(4 \cdot 9) \bmod 7 = 36 \bmod 7 = 1$.

My answer

Q63. [new]

Without a calculator, find $(52^{23} + 29 \cdot 86) \bmod 5$.

(A) 0

(B) 1

(C) 2

(D) 3

Answer: (C). Replace by remainders: $52 \equiv 2$, $29 \equiv 4$, $86 \equiv 1$. $2^4 = 16 \equiv 1$, so $2^{23} = (2^4)^5 2^3 \equiv 8 \equiv 3$. Total $3 + 4 = 7 \equiv 2$.

My answer

Q64. [new]

A hash table of size 11 uses the hash function $h(x) = (7x + 3) \bmod 11$. In which location is the record with ID $x = 245$ stored?

(A) 1

(B) 2

(C) 5

(D) 9

Answer: (B). $7 \cdot 245 + 3 = 1718 = 156 \cdot 11 + 2$.

My answer

Q65. [new]

Why is the hash function $h(x) = (ax + c) \bmod m$ suitable for a hash table with m locations?

(A) Its output is always in the range 1 through m.

(B) Its output is always in the range 0 through $m - 1$, so it can index the table directly.

(C) It is a one-to-one function on all ID numbers.

(D) It never produces the same location for two different IDs.

Answer: (B). Any value mod m lies in $\{0, \ldots, m - 1\}$. Collisions are possible because there are more IDs than locations.

My answer

Q66. [new]

A linear congruential generator uses $x_{i+1} = (5x_i + 3) \bmod 16$ with seed $x_0 = 7$. What is x_3?

(A) 6

(B) 1

(C) 11

(D) 8

Answer: (D). $x_1 = 38 \bmod 16 = 6$, $x_2 = 33 \bmod 16 = 1$, $x_3 = 8 \bmod 16 = 8$.

My answer

Q67. [new]

Suppose division in $\mathbb{Z}_6$ is defined by $x/y = z$ iff $y \cdot z = x$ in $\mathbb{Z}_6$. How many $z \in \mathbb{Z}_6$ satisfy $2 \cdot z = 4$?

(A) Exactly one, $z = 2$, so $4/2$ is well defined.

(B) None.

(C) Two, $z = 2$ and $z = 5$, so division is not well defined in $\mathbb{Z}_6$.

(D) Three, $z = 2$, $z = 5$ and $z = 8$.

Answer: (C). $2 \cdot 2 = 4$ and $2 \cdot 5 = 10 \equiv 4 \pmod 6$; $8 \notin \mathbb{Z}_6$.

My answer

Q68. [new]

If subtraction in $\mathbb{Z}_7$ is defined by $x - y = z$ iff $y + z = x$ in $\mathbb{Z}_7$, what is $2 - 5$ in $\mathbb{Z}_7$?

(A) -3

(B) 3

(C) 5

(D) 4

Answer: (D). $5 + 4 = 9 \equiv 2 \pmod 7$; also $-3 \bmod 7 = 4$.

My answer

Q69. [new]

In the multiplication table for $\mathbb{Z}_4$, the row for 2 (products $2 \cdot 0, 2 \cdot 1, 2 \cdot 2, 2 \cdot 3$) is

(A) $0, 2, 4, 6$

(B) $0, 2, 0, 2$

(C) $0, 2, 1, 3$

(D) $2, 0, 2, 0$

Answer: (B). $4 \bmod 4 = 0$ and $6 \bmod 4 = 2$.

My answer

Q70. [new]

Which $x \in \mathbb{Z}_5$ satisfies $3x = 1$ in $\mathbb{Z}_5$?

(A) 2

(B) 3

(C) 4

(D) No such x exists.

Answer: (A). $3 \cdot 2 = 6 \equiv 1 \pmod 5$.

My answer

Q71. [new]

Which set consists of integers that are all congruent to each other mod 4?

(A) $\{-7, 3, 9, 13\}$

(B) $\{-5, 1, 9, 13\}$

(C) $\{-1, 1, 9, 13\}$

(D) $\{-7, 1, 9, 13\}$

Answer: (D). Each has remainder 1 mod 4 ($-7 = (-2) \cdot 4 + 1$). 3, -5, -1 all have remainder 3.

My answer

Q72. [new]

The integers x with $-10 \le x \le 10$ and $x \equiv 3 \pmod 5$ are

(A) $\{-7, -2, 3, 8\}$

(B) $\{-8, -3, 3, 8\}$

(C) $\{-3, 3, 8\}$

(D) $\{-2, 3, 8\}$

Answer: (A). They are $3 + 5k$: $-7, -2, 3, 8$. Note $-3 \bmod 5 = 2$.

My answer

Q73. [new]

The algorithm "$p := 1$; For $i = 1$ to y: $p := (p \cdot x) \bmod n$; Return(p)" is run with $x = 3$, $y = 5$, $n = 7$. Which is the sequence of values of p after each iteration?

(A) $3, 9, 27, 81, 243$

(B) $3, 2, 6, 4, 5$

(C) $3, 2, 6, 4, 3$

(D) $3, 6, 5, 1, 3$

Answer: (B). 3; $9 \bmod 7 = 2$; 6; $18 \bmod 7 = 4$; $12 \bmod 7 = 5$. So $3^5 \bmod 7 = 5$.

My answer

Q74. [new]

Which identity holds for every integer $m > 1$ and all integers x, y?

(A) $(x + y) \bmod m = (x \bmod m) + (y \bmod m)$

(B) $(x \cdot y) \bmod m = (x \bmod m)(y \bmod m)$

(C) $(x + y) \bmod m = \big((x \bmod m) + (y \bmod m)\big) \bmod m$

(D) $(x - y) \bmod m = (x \bmod m) - (y \bmod m)$

Answer: (C). This is Theorem 5.3.2. Without the outer mod the others fail, e.g. $x = 4$, $y = 5$, $m = 6$: $9 \bmod 6 = 3 \neq 4 + 5$.

My answer

Q75. [new]

If $a \bmod 9 = 5$ and $b \bmod 9 = 7$, what is $(2a + 3b) \bmod 9$?

(A) 31

(B) 3

(C) 4

(D) 8

Answer: (C). $(2 \cdot 5 + 3 \cdot 7) \bmod 9 = 31 \bmod 9 = 4$.

My answer

Q76. [new]

Evaluate $9 \cdot 9 + 5$ in $\mathbb{Z}_{11}$.

(A) 86

(B) 4

(C) 9

(D) 3

Answer: (C). $9 \cdot 9 = 81 \equiv 4$, then $4 + 5 = 9$.

My answer

Q77. [new]

What is the binary representation of 45?

(A) 110101

(B) 101011

(C) 101101

(D) 100101

Answer: (C). $45 = 32 + 8 + 4 + 1$.

My answer

Q78. [new]

Since $11 = (1011)_2$, which product of powers of 7 (with exponents that are powers of 2) equals 7^{11}?

(A) $7^8 \cdot 7^4 \cdot 7^1$

(B) $7^8 \cdot 7^2 \cdot 7^1$

(C) $7^4 \cdot 7^2 \cdot 7^1$

(D) $7^8 \cdot 7^4 \cdot 7^2$

Answer: (B). $11 = 8 + 2 + 1$; the 2^2 bit is 0, so 7^4 is omitted.

My answer

Q79. [new]

Which expression is equivalent to x^{25}?

(A) $x^{16} \cdot x^8 \cdot x^2$

(B) $x^{16} \cdot x^4 \cdot x^1$

(C) $x^{16} \cdot x^8 \cdot x^4$

(D) $x^{16} \cdot x^8 \cdot x^1$

Answer: (D). $25 = (11001)_2 = 16 + 8 + 1$.

My answer

Q80. [new]

The binary representation of 44 is 101100. Which expression is equivalent to x^{44}?

(A) $x^{32} \cdot x^8 \cdot x^4$

(B) $x^{32} \cdot x^8 \cdot x^2$

(C) $x^{32} \cdot x^4 \cdot x^2$

(D) $x^{32} \cdot x^{16}$

Answer: (A). $44 = 32 + 8 + 4$.

My answer

Q81. [new]

How many times does the loop of the iterative fast exponentiation algorithm execute when the exponent is $y = 1000$?

(A) 9

(B) 10

(C) 11

(D) 1000

Answer: (B). The loop runs once per bit of y; $512 \le 1000 < 1024$, so $1000 = (1111101000)_2$ has 10 bits.

My answer

Q82. [new]

Computing x^y by repeatedly multiplying by x takes $y - 1$ multiplications. How many times does the loop of the iterative fast exponentiation algorithm execute?

(A) $y/2$ times

(B) $y - 1$ times

(C) The number of 1's in the binary representation of y

(D) The number of bits in the binary representation of y

Answer: (D). Each iteration removes the last bit ($r := r \operatorname{div} 2$), so there are $\lfloor \log_2 y \rfloor + 1$ iterations, i.e. $\Theta(\log y)$ operations.

My answer

Q83. [new]

The iterative fast exponentiation algorithm is: $p := 1$; $s := x$; $r := y$; While $(r > 0)$: If $(r \bmod 2 = 1)$ $p := p \cdot s$; $s := s \cdot s$; $r := r \operatorname{div} 2$; End-while; Return$(p)$. On input $x = 3$, $y = 13$, what are (p, s, r) at the beginning of the third iteration?

(A) $(3,\ 9,\ 6)$

(B) $(3,\ 81,\ 3)$

(C) $(27,\ 81,\ 3)$

(D) $(243,\ 6561,\ 1)$

Answer: (B). Start $(1, 3, 13)$; iteration 1 (r odd): $(3, 9, 6)$; iteration 2 (r even): $(3, 81, 3)$.

My answer

Q84. [new]

The fast exponentiation algorithm of the previous question is run with $y = 18$. In which iterations is p updated (i.e. $r \bmod 2 = 1$)?

(A) 1st and 4th

(B) 1st and 2nd

(C) 2nd and 5th

(D) 4th and 5th

Answer: (C). $18 = (10010)_2$; the bits are read from the right: $0, 1, 0, 0, 1$ ($r = 18, 9, 4, 2, 1$).

My answer

Q85. [new]

The fast modular exponentiation algorithm ($p := 1$; $s := x \bmod n$; $r := y$; While $(r > 0)$: If $(r \bmod 2 = 1)$ $p := (p \cdot s) \bmod n$; $s := (s \cdot s) \bmod n$; $r := r \operatorname{div} 2$) is run with $x = 3$, $y = 13$, $n = 7$. What is s at the beginning of the third iteration?

(A) 2

(B) 4

(C) 9

(D) 81

Answer: (B). s takes the values 3, $9 \bmod 7 = 2$, $4 \bmod 7 = 4$ at the beginning of iterations $1, 2, 3$.

My answer

Q86. [new]

What value does the fast modular exponentiation algorithm return on $x = 3$, $y = 13$, $n = 7$, i.e. what is $3^{13} \bmod 7$?

(A) 1

(B) 5

(C) 3

(D) 6

Answer: (C). $13 = 8 + 4 + 1$: $3^1 \equiv 3$, $3^4 \equiv 4$, $3^8 \equiv 2$, so $p = 3 \cdot 4 \cdot 2 = 24 \equiv 3 \pmod 7$.

My answer

Q87. [new]

Given $7^1 \bmod 13 = 7$, $7^2 \bmod 13 = 10$, $7^4 \bmod 13 = 9$, $7^8 \bmod 13 = 3$, $7^{16} \bmod 13 = 9$, compute $7^{21} \bmod 13$.

(A) 7

(B) 8

(C) 9

(D) 10

Answer: (B). $21 = 16 + 4 + 1$: $9 \cdot 9 \cdot 7 = 567 = 43 \cdot 13 + 8$.

My answer

Q88. [new]

Use the fact that $5^6 \bmod 7 = 1$ to find $5^{63} \bmod 7$.

(A) 1

(B) 5

(C) 4

(D) 6

Answer: (D). $5^{63} = (5^6)^{10} \cdot 5^3 \equiv 125 \equiv 6 \pmod 7$.

My answer

Q89. [new]

Without a calculator, find $2^{100} \bmod 7$.

(A) 1

(B) 4

(C) 2

(D) 6

Answer: (C). $2^3 = 8 \equiv 1$, so $2^{100} = (2^3)^{33} \cdot 2 \equiv 2 \pmod 7$.

My answer

Q90. [new]

Without a calculator, find $12^{37} \bmod 5$.

(A) 2

(B) 3

(C) 4

(D) 1

Answer: (A). $12 \equiv 2$ and $2^4 = 16 \equiv 1$, so $2^{37} = (2^4)^9 \cdot 2 \equiv 2 \pmod 5$.

My answer

Part B: Written problems

Q91.

Quiz 7 Alenzi

Let $a = 719$ and $n = 311$. (1) Use the Euclidean Algorithm to find $\gcd(a, n)$. (2) Use the Extended Euclidean Algorithm to write $\gcd(a, n)$ as a linear combination of a and n; that is, find integers s and t such that $\gcd(a, n) = a \cdot s + n \cdot t$.

Answer:

(1) Repeated division:

$$719 = 311 \cdot 2 + 97, \qquad 311 = 97 \cdot 3 + 20, \qquad 97 = 20 \cdot 4 + 17,$$
$$20 = 17 \cdot 1 + 3, \qquad 17 = 3 \cdot 5 + 2, \qquad 3 = 2 \cdot 1 + 1, \qquad 2 = 1 \cdot 2 + 0.$$

The last nonzero remainder is 1, so $\gcd(719, 311) = 1$.

(2) Rewrite each step as remainder $=$ dividend $-$ quotient $\cdot$ divisor and back-substitute, starting from $1 = 3 - 2$:

$$\begin{aligned} 1 &= 3 - (17 - 5 \cdot 3) = 6 \cdot 3 - 17 \\ &= 6(20 - 17) - 17 = 6 \cdot 20 - 7 \cdot 17 \\ &= 6 \cdot 20 - 7(97 - 4 \cdot 20) = 34 \cdot 20 - 7 \cdot 97 \\ &= 34(311 - 3 \cdot 97) - 7 \cdot 97 = 34 \cdot 311 - 109 \cdot 97 \\ &= 34 \cdot 311 - 109(719 - 2 \cdot 311) = (-109) \cdot 719 + 252 \cdot 311. \end{aligned}$$

So $s = -109$, $t = 252$. Check: $-78371 + 78372 = 1$. (Consequently $719^{-1} \equiv -109 \equiv 202 \pmod{311}$ and $311^{-1} \equiv 252 \pmod{719}$.)

My answer

Q92.

[new]

Prove or disprove each statement for all integers a, b, c (with $a \neq 0$, $b \neq 0$ wherever they appear as divisors): (a) If $a \mid b$ and $b \mid c$, then $a \mid c$. (b) If $a \mid b$ and $a \mid c$, then $a \mid (b + c)$. (c) If $a \mid bc$, then $a \mid b$ or $a \mid c$.

Answer:

(a) True. Since $a \mid b$, $b = ka$ for some integer k; since $b \mid c$, $c = jb$ for some integer j. Then $c = j(ka) = (jk)a$ with jk an integer, and $a \neq 0$, so $a \mid c$. ■

(b) True. $b = ka$ and $c = ja$ for integers k, j. Then $b + c = (k + j)a$ with $k + j$ an integer, so $a \mid (b + c)$. (This is the special case $s = t = 1$ of Theorem 5.1.1.) ■

(c) False. Counterexample: $a = 6$, $b = 2$, $c = 3$. Then $bc = 6$ and $6 \mid 6$, but $6 \nmid 2$ and $6 \nmid 3$.

My answer

Q93. [new]

(Uniqueness in the Division Algorithm.) Let n be an integer and d a positive integer. Suppose $n = q_1 d + r_1 = q_2 d + r_2$ where q_1, q_2, r_1, r_2 are integers with $0 \leq r_1 \leq d - 1$ and $0 \leq r_2 \leq d - 1$. Prove that $q_1 = q_2$ and $r_1 = r_2$.

Answer:

Subtracting the two expressions for n gives $0 = (q_1 - q_2)d + (r_1 - r_2)$, so

$$r_2 - r_1 = (q_1 - q_2)\, d.$$

Thus $r_2 - r_1$ is a multiple of d. From $0 \leq r_1, r_2 \leq d - 1$ we get $-(d - 1) \leq r_2 - r_1 \leq d - 1$, i.e. $|r_2 - r_1| < d$. The only multiple of d with absolute value less than d is 0, so $r_2 - r_1 = 0$, i.e. $r_1 = r_2$. Then $(q_1 - q_2)d = 0$ and since $d \neq 0$, $q_1 = q_2$. ■

My answer

Q94. **[new]**

Compute the following using the Division Algorithm, and in each case verify $n = qd + r$ with $0 \le r \le d - 1$: (a) $83 \operatorname{div} 12$ and $83 \bmod 12$; (b) $-47 \operatorname{div} 9$ and $-47 \bmod 9$, tracing the procedural algorithm; (c) $-60 \operatorname{div} 12$ and $-60 \bmod 12$. (d) For which integers x is $x \bmod 9 = 7$? List those in $[-20, 20]$.

Answer:

(a) Subtract 12 repeatedly: $83 \to 71 \to 59 \to 47 \to 35 \to 23 \to 11$; six subtractions, so $q = 6$, $r = 11$. Check: $6 \cdot 12 + 11 = 83$.

(b) $n < 0$, so add 9 until $r \ge 0$: $-47 \to -38 \to -29 \to -20 \to -11 \to -2 \to 7$; six additions, so $q = -6$, $r = 7$. Check: $(-6) \cdot 9 + 7 = -47$.

(c) $-60 = (-5) \cdot 12 + 0$, so $q = -5$, $r = 0$ (the loop stops as soon as $r = 0 \ge 0$).

(d) $x \bmod 9 = 7$ iff $x = 9k + 7$ for some integer k. In $[-20, 20]$: $-20, -11, -2, 7, 16$.

My answer

Q95. [new]

Use the extended Euclidean algorithm to find the multiplicative inverse of $43 \bmod 240$. Give the answer as an integer from 1 through 239 and check it.

Answer:

Euclid's algorithm:

$$240 = 5 \cdot 43 + 25, \qquad 43 = 1 \cdot 25 + 18, \qquad 25 = 1 \cdot 18 + 7,$$
$$18 = 2 \cdot 7 + 4, \qquad 7 = 1 \cdot 4 + 3, \qquad 4 = 1 \cdot 3 + 1, \qquad 3 = 3 \cdot 1 + 0.$$

So $\gcd(240, 43) = 1$ and an inverse exists. Back-substitution:

$$\begin{aligned} 1 &= 4 - 3 = 4 - (7 - 4) = 2 \cdot 4 - 7 \\ &= 2(18 - 2 \cdot 7) - 7 = 2 \cdot 18 - 5 \cdot 7 \\ &= 2 \cdot 18 - 5(25 - 18) = 7 \cdot 18 - 5 \cdot 25 \\ &= 7(43 - 25) - 5 \cdot 25 = 7 \cdot 43 - 12 \cdot 25 \\ &= 7 \cdot 43 - 12(240 - 5 \cdot 43) = 67 \cdot 43 - 12 \cdot 240. \end{aligned}$$

Hence $43 \cdot 67 \equiv 1 \pmod{240}$, and the inverse is 67. Check: $43 \cdot 67 = 2881 = 12 \cdot 240 + 1$.

My answer

Q96. [new]

Let x and y be positive integers and let d be a positive integer such that $d \mid x$, $d \mid y$, and $d = sx + ty$ for some integers s and t. Prove that $d = \gcd(x, y)$.

Answer:

Since d divides both x and y, d is a common divisor, so $d \le \gcd(x, y)$.

Let $g = \gcd(x, y)$. Then $g \mid x$ and $g \mid y$, so by Theorem 5.1.1 g divides every linear combination of x and y; in particular $g \mid sx + ty = d$. Since $d > 0$ and $g \mid d$, we have $g \le d$.

Combining, $d \le g \le d$, so $d = \gcd(x, y)$. ■

My answer

Q97. **[new]**

Let $m > 1$ be an integer. Prove that for all integers x and y, $x \equiv y \pmod{m}$ if and only if $m \mid (x - y)$.

Answer:

($\Rightarrow$) Suppose $x \equiv y \pmod{m}$, i.e. $x \bmod m = y \bmod m = r$. By the Division Algorithm, $x = km + r$ and $y = jm + r$ for some integers k, j. Then $x - y = (k - j)m$ with $k - j$ an integer, so $m \mid (x - y)$.

($\Leftarrow$) Suppose $m \mid (x - y)$, so $x - y = km$ for some integer k. Let $r = y \bmod m$; then $y = jm + r$ with $0 \leq r \leq m - 1$. Hence

$$x = y + km = (j + k)m + r.$$

Since $0 \leq r \leq m - 1$, uniqueness in the Division Algorithm gives $x \bmod m = r = y \bmod m$, so $x \equiv y \pmod{m}$. ■

My answer

Q98. **[new]**

(a) Give the addition and multiplication tables for $\mathbb{Z}_6$. (b) Define subtraction by $x - y = z$ iff $y + z = x$

in $\mathbb{Z}_6$. Is subtraction well defined (exactly one z for every pair x, y)? (c) Define division by $x/y = z$ iff $y \cdot z = x$ in $\mathbb{Z}_6$ ($y \neq 0$). Is division well defined? Which nonzero elements of $\mathbb{Z}_6$ have multiplicative inverses?

Answer:

(a)

$+$	0	1	2	3	4	5
0	0	1	2	3	4	5
1	1	2	3	4	5	0
2	2	3	4	5	0	1
3	3	4	5	0	1	2
4	4	5	0	1	2	3
5	5	0	1	2	3	4

$\times$	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	1	2	3	4	5
2	0	2	4	0	2	4
3	0	3	0	3	0	3
4	0	4	2	0	4	2
5	0	5	4	3	2	1

(b) Yes. Every row of the addition table contains each element of $\mathbb{Z}_6$ exactly once, so for each y and x there is exactly one z with $y + z = x$, namely $z = (x - y) \bmod 6$.

(c) No. In row 2 of the multiplication table, 4 appears twice ($2 \cdot 2 = 2 \cdot 5 = 4$), so $4/2$ is not unique, and 1 never appears, so $1/2$ does not exist. Only the rows of 1 and 5 contain every element; the elements with inverses are 1 ($1 \cdot 1 = 1$) and 5 ($5 \cdot 5 = 25 \equiv 1$), exactly those relatively prime to 6.

My answer

Q99. [new]

Let $m > 1$ and let x, y be integers. Prove that

$$(x + y) \bmod m = \big((x \bmod m) + (y \bmod m)\big) \bmod m \quad \text{and}$$
$$(xy) \bmod m = \big((x \bmod m)(y \bmod m)\big) \bmod m.$$

Then use this to compute $(43^{17} + 32 \cdot 139) \bmod 7$ without a calculator.

Answer:

By the Division Algorithm, $x = km + a$ and $y = jm + b$ where $a = x \bmod m$, $b = y \bmod m$, and k, j are integers.

Sum: $x + y = (a + b) + (k + j)m$. So $x + y$ and $a + b$ differ by a multiple of m, hence $m \mid (x + y) - (a + b)$ and they have the same remainder mod m: $(x + y) \bmod m = (a + b) \bmod m$.

Product: $xy = (km + a)(jm + b) = ab + (kjm + kb + aj)m$. So $xy - ab$ is a multiple of m, hence $(xy) \bmod m = (ab) \bmod m$. ■

Application: $43 \bmod 7 = 1$, $32 \bmod 7 = 4$, $139 \bmod 7 = 6$. Applying the identities repeatedly (a power is a repeated product),

$$(43^{17} + 32 \cdot 139) \bmod 7 = (1^{17} + 4 \cdot 6) \bmod 7 = 25 \bmod 7 = 4.$$

My answer

Q100. **[new]**

Use the iterative fast modular exponentiation algorithm to compute $6^{45} \bmod 29$. Show the binary expansion of 45 and the values of p, s, r at the beginning of each iteration.

Answer:

$45 = (101101)_2 = 32 + 8 + 4 + 1$. Algorithm: $p := 1$; $s := 6$; $r := 45$; while $r > 0$: if $r \bmod 2 = 1$ then $p := ps \bmod 29$; $s := s^2 \bmod 29$; $r := r \operatorname{div} 2$.

iteration	p	s	r	$r \bmod 2$	update
1	1	6	45	1	$p = 6$, $s = 36 \bmod 29 = 7$
2	6	7	22	0	$s = 49 \bmod 29 = 20$
3	6	20	11	1	$p = 120 \bmod 29 = 4$, $s = 400 \bmod 29 = 23$
4	4	23	5	1	$p = 92 \bmod 29 = 5$, $s = 529 \bmod 29 = 7$
5	5	7	2	0	$s = 49 \bmod 29 = 20$
6	5	20	1	1	$p = 100 \bmod 29 = 13$, $s = 23$
end	13	23	0		loop stops

So $6^{45} \bmod 29 = 13$. The loop ran 6 times, one per bit of 45, using about $2 \cdot 6$ multiplications instead of 44.

My answer